Ibrahim Hussain

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Quantitative Data Scientist | AI Systems, Data Engineering & Financial Risk Modelling

Quantitative Data Scientist focused on statistical modelling, time-series forecasting, and economic risk analysis. Skilled in time-series modelling, and multi-source data fusion, with a strong focus on extracting actionable insights from complex, high-dimensional datasets. Built and deployed 10+ projects including multi-agent AI systems, large-scale economic modelling, and multi-omics data integration, combining modern & traditional AI frameworks to solve real-world problems.

Experience

SMARTFORUM

AI Engineer Intern *June – September 2025*

- Optimised RAG pipelines using PostgreSQL indexing and query optimisation, reducing retrieval latency by ~20% and improving response accuracy.
- Engineered modular multi-agent AI workflows using LangChain and LangGraph, automating core data-processing tasks and reducing manual intervention.
- Designed structured database schemas for efficient data storage, retrieval, and agent coordination in production workflows.

Projects & Technical Experience

EXPLAINABLE FINANCIAL RISK MANAGEMENT FRAMEWORK

AI Systems & Data Engineering Lead *January – April 2026*

- Led end-to-end design and implementation of an explainable multimodal deep learning framework for financial risk management, combining market, SEC filings, macroeconomic, and graph-based data streams.
- Engineered large-scale financial data pipelines across 25 years of U.S. market data, including a final cleaned panel of 2,500 stocks × 6,286 trading days with zero missing values after multi-source filling and feature engineering.
- Processed SEC EDGAR fundamentals and textual filings, including 900k+ submissions JSON files, 17GB+ company facts, and 300k+ raw SEC filing documents.
- Designed the full model architecture and selection rationale, including a shared temporal attention encoder, FinBERT text encoder, sentiment/news analysts, StemGNN contagion module, regime risk module, drawdown module, liquidity risk module, and fusion layers.
- Adapted baseline graph neural network research models, including StemGNN and MTGNN, from price forecasting into specialised financial risk modules for contagion detection and market-regime identification.
- Built 3 leakage-safe chronological training splits across 2000–2024, using train/validation/test chunks to preserve temporal validity and improve model defensibility.
- Produced comprehensive technical documentation covering data engineering methodology, model-selection rationale, evaluation metrics, XAI design, and reproducible workflow decisions.

INFLATION DYNAMICS IN PAKISTAN

Statistical Researcher *May 2025*

- Analysed 43 years (1980-2023) of macroeconomic data to model inflation dynamics.
- Built and compared ARIMA, Ridge, Lasso, and Elastic Net models using AIC, RMSE, and MSE for model selection.
- Conducted hypothesis testing and correlation analysis to identify exchange rate volatility and money supply growth as statistically significant drivers of inflation.
- Engineered features and handled missing data through structured imputation and validation techniques.

DISASTER RECOVERY ANALYSIS (RRS)

Data Engineer *December 2025*

- Engineered a multi-source dataset integrating disaster records, GDP growth, governance indices, and HDI across 7,000+ observations (1998–2023).

- Designed the Resilience Recovery Score by modelling pre and post-disaster GDP dynamics and recovery timelines.
- Performed extensive feature engineering, and relational joins across macroeconomic and demographic datasets.
- Built interactive visual analytics to analyse global resilience patterns and recovery behaviour.

Education

FAST UNIVERSITY Islamabad, Pakistan
Bachelor of Science in Data Science *September 2023 – Present*
Relevant Coursework: Machine Learning, Deep Learning, Data Mining, Big Data Systems, Database Systems & Warehousing, Advanced Statistics, Time Series Analysis.

Technical Skills

Programming: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL (PostgreSQL, MySQL), R.
Quantitative Modelling: Financial Risk Modelling, Volatility, Drawdown Analysis, Portfolio Metrics, Regime Detection, Contagion Risk.
Data Engineering: Large-Scale Data Pipelines, Time-Series Data Engineering, Data Cleaning, Missing Value Imputation, Multi-Source Data Fusion, Feature Engineering.
Data Science & Big Data Modelling: Time-Series Analysis, Regression (Ridge, Lasso, Elastic Net), Hypothesis Testing, Feature Engineering.
Machine Learning & AI: Deep Learning, Transformers, LangChain, LangGraph, Graph Neural Networks, Multi-Agent Systems.
Interests: Economic Analysis, Machine Learning Algorithms, Economic Modelling, Digital Marketing Analytics.
Language: English (Fluent), Urdu (Native), Italian (Basic).

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